Distance on the Coordinate Plane

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 5 — Distance on the Coordinate Plane	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.2

2. Standards

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must. Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

Must not. The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.2.1	Demonstrate understanding by representing problems in multiple ways.	Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 3 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	Notes I and board 9: a distance across zero checked by counting the squares

4. Learning Target and Essential Question

Learning target. I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

Essential question. How can two pairs of numbers tell you how far apart two points are?

Building on. 6.03–04 — the Pythagorean Theorem for a missing side; Grade 6 (MA.6.GR.1.2) — the distance between points with the same x or the same y; Grade 7 — subtracting integers.

Working toward. 6.06 — perimeters and maps on the grid; Algebra 1 and Geometry — the distance formula, which is this triangle written as one line.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front; say the off-slide sentence out loud.	Copy the notes; answer the checks aloud.
Example 1	3	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; due by the Unit 6 test.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the off-slide sentence is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
Why is the distance from (−4, 3) to (5, 3) not 5 − 4?	DOK 2
Two points have different x's and different y's. Why can you not just subtract?	DOK 3
Does it matter which point you start from? Why does the answer come out the same?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	How far apart are $(-7, -4)$ and $(-7, 6)$?	***10 units***	2 — subtracted 6 − 4 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]
2	How far apart are $(-9, 2)$ and $(3, 2)$?	***12 units***	6 — subtracted 9 − 3 across zero [Florida B1G-M, GR.1.2: integer errors across an axis]
3	What is the distance between (−5, 1) and (3, 1)?	***B — 8 units***	(A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Florida B1G-M, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I]
4	How far apart are $A(-4, -3)$ and $B(5, 9)$?	***15 units***	21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn']
5	How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth?	***$\sqrt{89} \approx 9.4$ units***	13 — added the legs [MA.8.GR.1.2]; $\sqrt{(1^2 + 8^2)}$ — subtracted 3 − 2 across zero [Florida B1G-M, GR.1.2]
6	Which gives the distance between (4, 5) and (−3, −1)?	***C — $\sqrt{(7^2 + 6^2)} = \sqrt{85}$***	(A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Florida B1G-M, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum]
7	How far is $(5, -12)$ from the origin?	***13 units***	17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares]
8	On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$. How far is the school from the library?	***7 mi***	1 — subtracted 4 − 3 across zero [Florida B1G-M, GR.1.2]; a distance to the town hall — the wrong pair of points
9	Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance?	***He subtracted 7 − 3. Across zero the pieces add: 10 units.***	'4 is right' — the axis-crossing error [Florida B1G-M, GR.1.2]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. A printed grid for every board. The dashed legs are drawn on Notes II–III and the examples; from board 4 the student draws them. "Across, then up or down" said aloud with every slanted board.

ELL. **Horizontal**: side to side, like the horizon. **Vertical**: up and down. **Distance**: how far apart — never negative. Point at the grid with each word.

Enrichment. Find every point on the grid exactly 5 units from the origin with whole-number coordinates. (There are 12: (±5, 0), (0, ±5), (±3, ±4), (±4, ±3).)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: Ty subtracted 7 − 3, but −3 and 7 are on opposite sides of zero, so the two pieces add: 3 + 7 = 10. A board that says only '10' without naming Ty's step has the answer and not the error.

Vocabulary. **horizontal distance**; **vertical distance**; **legs on the grid**

Materials. Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 6 test: Find the distance between two points on a horizontal or vertical line — L9V; Find the distance between two points — ZBP.